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## MIDTERM Preparation — Solutions

Thursday, May 7th 2026

This exam has 4 problems for 100 points in total + 10 bonus points.

The exam time is 1 hour 50 minutes.

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### Instructions

- The exam is closed book, and you are allowed **one page of pre-prepared reference material**. You can quote results from the lecture notes, HWs, discussion problems and extra problems as part of your solutions to the exam problems; but this has to be quite precise.
- It is your responsibility to **number the extra pages** of your solutions and write on the first sheet the **total number of extra pages** submitted.
- You **cannot** collaborate in any way with any individual on the exam. Any form of communication/consultation/collaboration with another person about the exam is expressly prohibited. Violation of the no-collaboration policy is a violation of the UCLA code of student conduct and will come with serious consequences.
- Please solve every problem on the blank pages; if you need extra pages, add to them in the same order and number them, please make sure to clarify which problem is being attempted. On each such extra page, please write down the problem you are solving, your name, and UCLA UID.
- **All answers must be justified. Mysterious or unsupported answers will not receive credit.** A correct answer, unsupported by calculations and/or explanation will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- By putting your name on the exam and submitting it, you are acknowledging that you are adhering to the honor code and rules of the exam given.

NAME: \_\_\_\_\_

UID: \_\_\_\_\_

EXTRA PAGES: \_\_\_\_\_

| Problem | 1          | 2  | 3  | 4          | TOTAL        |
|---------|------------|----|----|------------|--------------|
| Max.    | 15+5 bonus | 25 | 30 | 30+5 bonus | 100+10 bonus |
| Awarded |            |    |    |            |              |

## Some useful relations and definitions

Please note that you may not require all of the following for the exam.

- The sign function is defined as

$$\text{sign}(a) = \begin{cases} 1 & ; a > 0 \\ -1 & ; a \leq 0 \end{cases}$$

- The sigmoid function is defined as  $\sigma(x) = \frac{1}{1+e^{-x}}$ . The derivative of the sigmoid function is  $\frac{e^{-x}}{(1+e^{-x})^2}$ .

## Problem 1 (TRUE/FALSE (15 pts + 5 bonus pts))

Please justify your answers. A correct answer with incorrect or absent explanation will not get any credit.

(a) **Logistic Regression:**

- i. [7 points] The prediction of a logistic regression classifier is given by  $y = \sigma(\mathbf{w}^\top \mathbf{x}) \in [0, 1]$ . Moreover, the decision boundary of this classifier is  $\mathbf{w}^\top \mathbf{x} \geq 0$ .

**TRUE / FALSE.**

**Solution: True.** The sigmoid output lies in  $[0, 1]$  and is thresholded at  $\frac{1}{2}$  for classification. Since  $\sigma(t) \geq \frac{1}{2}$  if and only if  $t \geq 0$ , the corresponding decision rule is  $\mathbf{w}^\top \mathbf{x} \geq 0$ . Equivalently, the boundary  $\sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{2}$  corresponds to  $\mathbf{w}^\top \mathbf{x} = 0$ .

- ii. [8 points] Let  $h_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$ ,  $(\mathbf{x}, y)$  a sample, and  $\sigma(a) = \frac{1}{1+e^{-a}}$ . When  $y = 1$ , it is true that

$$-y \log(\sigma(h_{\mathbf{w}}(\mathbf{x}))) - (1 - y) \log(1 - \sigma(h_{\mathbf{w}}(\mathbf{x}))) = \log(1 + e^{h_{\mathbf{w}}(\mathbf{x})}) - h_{\mathbf{w}}(\mathbf{x}) \quad (1)$$

**TRUE / FALSE.**

**Solution: True.** When  $y = 1$ , the  $(1 - y)$  term vanishes, so the LHS reduces to  $-\log \sigma(h_{\mathbf{w}}(\mathbf{x}))$ . Compute:

$$\begin{aligned} -\log \sigma(h_{\mathbf{w}}(\mathbf{x})) &= -\log\left(\frac{1}{1 + e^{-h_{\mathbf{w}}(\mathbf{x})}}\right) \\ &= \log(1 + e^{-h_{\mathbf{w}}(\mathbf{x})}) \\ &= \log\left(\frac{e^{h_{\mathbf{w}}(\mathbf{x})} + 1}{e^{h_{\mathbf{w}}(\mathbf{x})}}\right) \\ &= \log(1 + e^{h_{\mathbf{w}}(\mathbf{x})}) - h_{\mathbf{w}}(\mathbf{x}), \end{aligned}$$

which equals the RHS.

(b) **Linear regression:**

- i. **[bonus][2 points]** A closed form solution for linear regression is only possible if  $\mathbf{X}^\top \mathbf{X}$  is positive definite. Note that a matrix  $\mathbf{A}$  is positive definite if  $\mathbf{z}^\top \mathbf{A} \mathbf{z} > 0$  for all vectors  $\mathbf{z} \neq \mathbf{0}$ .

**TRUE / FALSE.**

*Hint: A matrix is positive definite if and only if all the eigenvalues are positive. Moreover, the determinant of any matrix  $\mathbf{A}$  is always equal to the product of all eigenvalues of  $\mathbf{A}$ .*

**Solution: False.** The formula  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  does require  $\mathbf{X}^\top \mathbf{X}$  to be invertible (equivalently, positive definite, since  $\mathbf{X}^\top \mathbf{X}$  is always positive semidefinite). However, even when  $\mathbf{X}^\top \mathbf{X}$  is singular, a closed-form minimizer of the least-squares objective exists via the Moore–Penrose pseudoinverse:  $\boldsymbol{\theta}^* = \mathbf{X}^+ \mathbf{y}$ .

(c) **Bias-variance trade-off:**

- i. **[bonus][3 points]** Regularized solution minimizing  $J(\mathbf{w}) + \lambda R(\mathbf{w})$  leads to a model with a higher empirical training loss  $J(\mathbf{w})$ .

**TRUE / FALSE.**

**Solution: False as stated.** Compared with an unregularized minimizer of  $J$ , the regularized solution cannot have *smaller* empirical loss, but it need not be *strictly* higher. Equality occurs, for example, when an unregularized minimizer also happens to minimize  $R$  (e.g.,  $\mathbf{w} = \mathbf{0}$  when it is already optimal), or trivially when  $\lambda = 0$ . The correct statement is that the empirical loss is at least as large, not strictly larger.

**Problem 2** (WEIGHTED LEAST SQUARES (25 pts))

In class you have seen linear regression where the objective is as follows

$$J(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2, \quad \mathbf{X} \in \mathbb{R}^{n \times d}, \boldsymbol{\theta} \in \mathbb{R}^d, \mathbf{y} \in \mathbb{R}^n \quad (2)$$

where the rows of  $\mathbf{X}$  are the data points,  $d$  is the dimension of the data points, and  $n$  is the number of data points. We had seen that the closed form for the global minimizer of  $J(\boldsymbol{\theta})$  is

$$\boldsymbol{\theta}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (3)$$

if  $\mathbf{X}^T \mathbf{X}$  is invertible.

Now consider the weighted least squares objective function  $\tilde{J}(\boldsymbol{\theta})$ :

$$\tilde{J}(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_{\mathbf{W}}^2 \quad (4)$$

where  $\mathbf{W} \in \mathbb{R}^{n \times n}$  is a diagonal matrix, i.e.

$$\mathbf{W} = \text{diag}(w_1, \dots, w_n) = \begin{bmatrix} w_1 & 0 & 0 & 0 \\ 0 & w_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & w_n \end{bmatrix} \quad (5)$$

with each diagonal element  $w_i > 0$  and

$$\|\mathbf{z}\|_{\mathbf{W}}^2 = \mathbf{z}^T \mathbf{W} \mathbf{z} \quad (6)$$

for any vector  $\mathbf{z} \in \mathbb{R}^n$ .

- (a) [11 points] Find the gradient of  $\tilde{J}(\boldsymbol{\theta})$ .

**Solution:** Since  $\mathbf{W}$  is diagonal, it is symmetric. Expanding the objective,

$$\begin{aligned} \tilde{J}(\boldsymbol{\theta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T \mathbf{W} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \\ &= \mathbf{y}^T \mathbf{W} \mathbf{y} - 2\mathbf{y}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{X}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta}. \end{aligned}$$

Taking the gradient with respect to  $\boldsymbol{\theta}$ ,

$$\nabla_{\boldsymbol{\theta}} \tilde{J}(\boldsymbol{\theta}) = -2\mathbf{X}^T \mathbf{W} \mathbf{y} + 2\mathbf{X}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta} = 2\mathbf{X}^T \mathbf{W} (\mathbf{X} \boldsymbol{\theta} - \mathbf{y}).$$

- (b) [9 points] Suppose that  $\mathbf{X}^T \mathbf{W} \mathbf{X}$  is invertible. Find the optimal solution  $\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \tilde{J}(\boldsymbol{\theta})$ .

**Solution:** Setting the gradient equal to zero,

$$2\mathbf{X}^T \mathbf{W} (\mathbf{X} \hat{\boldsymbol{\theta}} - \mathbf{y}) = \mathbf{0} \implies \mathbf{X}^T \mathbf{W} \mathbf{X} \hat{\boldsymbol{\theta}} = \mathbf{X}^T \mathbf{W} \mathbf{y}.$$

Since  $\mathbf{X}^T \mathbf{W} \mathbf{X}$  is invertible,

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}.$$

This is a global minimum because  $\tilde{J}$  is convex: its Hessian  $2\mathbf{X}^T \mathbf{W} \mathbf{X}$  is positive semidefinite (each  $w_i > 0$ ).

(c) [5 points] Compare  $\tilde{J}(\boldsymbol{\theta})$  to

$$J(\boldsymbol{\theta}) = \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 = \sum_{i=1}^n \left(y_i - \mathbf{x}_i^\top \boldsymbol{\theta}\right)^2. \quad (7)$$

How does the objective  $\tilde{J}(\boldsymbol{\theta})$  relate to  $J(\boldsymbol{\theta})$ ? What is the role of each  $w_i$ ?

**Solution:** Because  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ ,

$$\tilde{J}(\boldsymbol{\theta}) = \sum_{i=1}^n w_i \left(y_i - \mathbf{x}_i^\top \boldsymbol{\theta}\right)^2.$$

Thus  $\tilde{J}$  is a *weighted* version of ordinary least squares: each squared residual is scaled by its own weight. A larger  $w_i$  makes the error on data point  $i$  matter more, while a smaller  $w_i$  down-weights it. Setting all  $w_i = 1$  recovers  $J$ .

### Problem 3 (PROVE OR DISPROVE VALIDITY OF KERNELS (30pts))

State [YES/NO] for the following being valid positive semi-definite kernels or not. To receive full credit, you must provide an associated linear mapping for the [YES] case, or construct a counter-example for the [NO] case. No points would be awarded for answers without justification.

- (a) [7 points] Let  $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$  and consider

$$k(\mathbf{x}, \mathbf{x}') = \frac{1}{\mathbf{x}^T \mathbf{x}'}$$

Is  $k(\cdot, \cdot)$  a valid kernel?

**Solution: No.** First, the expression is undefined whenever  $\mathbf{x}^T \mathbf{x}' = 0$  (e.g., orthogonal points), so  $k$  is not even well-defined on all of  $\mathbb{R}^n \times \mathbb{R}^n$ . Even restricting to inputs where it is defined, the Gram matrix need not be PSD.

Take  $\mathbf{x}_1 = (1, 0)^T$  and  $\mathbf{x}_2 = (1, 1)^T$ . Then  $\mathbf{x}_1^T \mathbf{x}_1 = 1$ ,  $\mathbf{x}_2^T \mathbf{x}_2 = 2$ ,  $\mathbf{x}_1^T \mathbf{x}_2 = 1$ , giving

$$\mathbf{K} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \det(\mathbf{K}) = \frac{1}{2} - 1 = -\frac{1}{2} < 0.$$

A negative determinant of a  $2 \times 2$  symmetric matrix implies a negative eigenvalue, so  $\mathbf{K}$  is not positive semidefinite. Hence  $k$  is not a valid kernel.

- (b) [7 points] Consider

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{A} \mathbf{x}',$$

where  $\mathbf{A}$  is a positive definite matrix. Is  $k(\cdot, \cdot)$  a valid kernel?

*Hint: Consider the eigendecomposition of  $\mathbf{A}$ ,  $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$ . You may also find the following definitions/facts useful:  $\mathbf{A}^{\frac{1}{2}} := \mathbf{Q} \mathbf{\Lambda}^{\frac{1}{2}} \mathbf{Q}^T$ ,  $\mathbf{A} = \mathbf{A}^{\frac{1}{2}} \mathbf{A}^{\frac{1}{2}}$ , and  $\mathbf{A}^{\frac{1}{2}}$  is symmetric by definition.*

**Solution: Yes.** Since  $\mathbf{A}$  is positive definite, its eigenvalues are positive and  $\mathbf{A}^{1/2} = \mathbf{Q} \mathbf{\Lambda}^{1/2} \mathbf{Q}^T$  is well-defined and symmetric, with  $\mathbf{A} = \mathbf{A}^{1/2} \mathbf{A}^{1/2}$ . Define the feature map

$$\phi(\mathbf{x}) = \mathbf{A}^{1/2} \mathbf{x}.$$

Then

$$\phi(\mathbf{x})^T \phi(\mathbf{x}') = \mathbf{x}^T (\mathbf{A}^{1/2})^T \mathbf{A}^{1/2} \mathbf{x}' = \mathbf{x}^T \mathbf{A}^{1/2} \mathbf{A}^{1/2} \mathbf{x}' = \mathbf{x}^T \mathbf{A} \mathbf{x}',$$

so  $k$  admits a valid feature-map representation and is therefore a valid kernel.

(c) [8 points] Consider

$$k(\mathbf{x}, \mathbf{x}') = e^{-\|\mathbf{x} - \mathbf{x}'\|_2^2 + \ln(\mathbf{x}^\top \mathbf{x}')}.$$

where  $\mathbf{x}, \mathbf{x}' \in \mathbb{R}_{++}^n$ . Is  $k(\cdot, \cdot)$  a valid kernel?

**Solution: Yes.** Since  $\mathbf{x}, \mathbf{x}' \in \mathbb{R}_{++}^n$  (strictly positive entries),  $\mathbf{x}^\top \mathbf{x}' > 0$  and  $\ln(\mathbf{x}^\top \mathbf{x}')$  is well-defined. Splitting the exponent,

$$k(\mathbf{x}, \mathbf{x}') = e^{-\|\mathbf{x} - \mathbf{x}'\|_2^2} e^{\ln(\mathbf{x}^\top \mathbf{x}')} = \underbrace{e^{-\|\mathbf{x} - \mathbf{x}'\|_2^2}}_{k_R(\mathbf{x}, \mathbf{x}')} \cdot \underbrace{(\mathbf{x}^\top \mathbf{x}')}_{k_L(\mathbf{x}, \mathbf{x}')}.$$

Both factors are valid kernels:  $k_R$  is the Gaussian/RBF kernel (a known PSD kernel), and  $k_L$  is the linear kernel with feature map  $\phi_L(\mathbf{x}) = \mathbf{x}$ .

By the *Schur product theorem*, the Hadamard (entrywise) product of two PSD matrices is PSD; hence the product of two valid kernels is a valid kernel. Equivalently, an explicit feature map is the tensor product

$$\phi(\mathbf{x}) = \phi_R(\mathbf{x}) \otimes \mathbf{x}, \quad \phi(\mathbf{x})^\top \phi(\mathbf{x}') = (\phi_R(\mathbf{x})^\top \phi_R(\mathbf{x}'))(\mathbf{x}^\top \mathbf{x}') = k(\mathbf{x}, \mathbf{x}').$$

Therefore  $k$  is a valid kernel.

(d) [8 points] Let  $k_1(\cdot, \cdot)$  and  $k_2(\cdot, \cdot)$  be two valid kernels. Consider

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}'). \quad (8)$$

Prove that  $k(\cdot, \cdot)$  is a valid kernel.

**Solution:** Since  $k_1$  and  $k_2$  are valid kernels, there exist feature maps  $\phi_1$  and  $\phi_2$  such that

$$k_1(\mathbf{x}, \mathbf{x}') = \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}'), \quad k_2(\mathbf{x}, \mathbf{x}') = \phi_2(\mathbf{x})^\top \phi_2(\mathbf{x}').$$

Define the concatenated feature map

$$\phi(\mathbf{x}) = \begin{bmatrix} \phi_1(\mathbf{x}) \\ \phi_2(\mathbf{x}) \end{bmatrix}.$$

Then

$$\phi(\mathbf{x})^\top \phi(\mathbf{x}') = \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}') + \phi_2(\mathbf{x})^\top \phi_2(\mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}'),$$

so  $k$  admits a feature-map representation and is therefore valid.

*Equivalent argument via Gram matrices:* for any points  $\{\mathbf{x}_i\}$ , the Gram matrix is  $\mathbf{K} = \mathbf{K}_1 + \mathbf{K}_2$ , and for any  $\mathbf{z}$ ,

$$\mathbf{z}^\top \mathbf{K} \mathbf{z} = \mathbf{z}^\top \mathbf{K}_1 \mathbf{z} + \mathbf{z}^\top \mathbf{K}_2 \mathbf{z} \geq 0,$$

since  $\mathbf{K}_1, \mathbf{K}_2$  are PSD.

**Problem 4** (SVM WITH MODIFIED COST (30 pts + 5 pts bonus))

Consider a binary classification problem where you are given a dataset  $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ , with  $\mathbf{x}_i \in \mathbb{R}^d$  and  $y_i \in \{-1, 1\}$ . We consider a variant of the classical SVM problem done in class. The primal form of the new soft-margin SVM optimization problem is:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^n \left( C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right) \right) \xi_i \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \end{aligned} \tag{9}$$

where  $C_1 > 0$  and  $C_{-1} > 0$  are fixed hyperparameters, *i.e.*, we do not optimize over them.

- (a) [3 points] Compare (9) with the vanilla soft-margin SVM problem covered in class.

*Hint: Consider the cases when  $y_i = 1$  and  $y_i = -1$ .*

**Solution:** In the vanilla soft-margin SVM, every slack  $\xi_i$  has the same coefficient  $C$ . Here the coefficient depends on the label  $y_i \in \{-1, +1\}$ :

- If  $y_i = +1$ :  $C_{-1} \left( \frac{1-1}{2} \right) + C_1 \left( \frac{1+1}{2} \right) = C_1$ .
- If  $y_i = -1$ :  $C_{-1} \left( \frac{1-(-1)}{2} \right) + C_1 \left( \frac{1+(-1)}{2} \right) = C_{-1}$ .

So margin violations on positive examples are penalized by  $C_1$  and on negative examples by  $C_{-1}$ . The two classes can carry different misclassification costs.

- (b) [5 points] Write down the Lagrangian function for the primal problem, incorporating the constraints with Lagrange multipliers  $\alpha_i$  and  $\lambda_i$ . Introduce the dual variables  $\alpha_i$  for the constraints  $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i$  and  $\lambda_i$  for the constraints  $\xi_i \geq 0$ . Show that the Lagrangian is equal to:

$$L(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\lambda}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^n \left( C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right) \right) \xi_i - \sum_{i=1}^n \alpha_i \left[ y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 + \xi_i \right] - \sum_{i=1}^n \lambda_i \xi_i. \quad (10)$$

where the vectors  $\boldsymbol{\alpha}$ ,  $\boldsymbol{\lambda}$ , and  $\boldsymbol{\xi}$  are defined as follows:

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}, \quad \boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}.$$

**Solution:** Rewrite the constraints in the standard  $\geq 0$  form:

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 + \xi_i \geq 0, \quad \xi_i \geq 0.$$

For a minimization problem with constraints  $h_i(\cdot) \geq 0$  and nonnegative multipliers  $\alpha_i, \lambda_i \geq 0$ , the Lagrangian subtracts  $\sum \alpha_i h_i$ . This gives directly

$$L(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\lambda}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^n \left( C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right) \right) \xi_i - \sum_{i=1}^n \alpha_i \left[ y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 + \xi_i \right] - \sum_{i=1}^n \lambda_i \xi_i.$$

- (c) Derive the dual problem by taking partial derivatives of the Lagrangian with respect to the primal variables  $\mathbf{w}$ ,  $b$ , and  $\xi_i$ , and setting them to zero.

- (i) [2 points] Find the gradient of the Lagrangian with respect to  $\mathbf{w}$  and set it to zero.

**Solution:**

$$\nabla_{\mathbf{w}} L = \mathbf{w} - \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i = \mathbf{0} \implies \boxed{\mathbf{w} = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i.}$$

- (ii) [2 points] Find the partial derivative of the Lagrangian with respect to  $b$  and set it to zero.

**Solution:**

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^n \alpha_i y_i = 0 \implies \boxed{\sum_{i=1}^n \alpha_i y_i = 0.}$$

- (iii) [3 points] Find the partial derivative of the Lagrangian with respect to  $\xi_i$  and set it to zero.

**Solution:** For brevity let  $C_i := C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right)$ , i.e.,  $C_i = C_1$  if  $y_i = 1$  and  $C_i = C_{-1}$  if  $y_i = -1$ . Then

$$\frac{\partial L}{\partial \xi_i} = C_i - \alpha_i - \lambda_i = 0 \implies \alpha_i + \lambda_i = C_i.$$

Since  $\alpha_i \geq 0$  and  $\lambda_i \geq 0$ ,

$$\boxed{0 \leq \alpha_i \leq C_i = C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right).}$$

- (d) [6 points] Substitute the expressions obtained from the partial derivatives back into the Lagrangian. Prove the simplified dual formulation is equal to the following equation:

$$g(\boldsymbol{\alpha}) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\top \mathbf{x}_j) + \sum_{i=1}^n \alpha_i. \quad (11)$$

**Solution:** Group the Lagrangian by primal variable:

$$L = \frac{1}{2} \|\mathbf{w}\|^2 - \mathbf{w}^\top \left( \sum_i \alpha_i y_i \mathbf{x}_i \right) - b \left( \sum_i \alpha_i y_i \right) + \sum_i (C_i - \alpha_i - \lambda_i) \xi_i + \sum_i \alpha_i.$$

Apply the stationarity conditions:  $\sum_i \alpha_i y_i = 0$  kills the  $b$ -term,  $C_i - \alpha_i - \lambda_i = 0$  kills the  $\xi_i$ -terms, and  $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$  gives

$$\frac{1}{2} \|\mathbf{w}\|^2 - \mathbf{w}^\top \mathbf{w} = -\frac{1}{2} \|\mathbf{w}\|^2.$$

Therefore

$$\begin{aligned} g(\boldsymbol{\alpha}) &= -\frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^n \alpha_i \\ &= -\frac{1}{2} \left\| \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i \right\|^2 + \sum_{i=1}^n \alpha_i \\ &= -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\top \mathbf{x}_j) + \sum_{i=1}^n \alpha_i. \end{aligned}$$

- (e) [5 points] Write down the final dual optimization problem, including the constraints.

**Solution:**

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\top \mathbf{x}_j) \\ \text{s.t.} \quad & \sum_{i=1}^n \alpha_i y_i = 0, \\ & 0 \leq \alpha_i \leq C_{-1} \left( \frac{1-y_i}{2} \right) + C_1 \left( \frac{1+y_i}{2} \right), \quad i = 1, \dots, n. \end{aligned}$$

Equivalently:  $0 \leq \alpha_i \leq C_1$  for positive examples ( $y_i = +1$ ) and  $0 \leq \alpha_i \leq C_{-1}$  for negative examples ( $y_i = -1$ ).

- (f) [4 points] Is this problem kernelizable?

**Solution: Yes.** The data points appear in the dual objective only through inner products  $\mathbf{x}_i^\top \mathbf{x}_j$ . Replacing each such inner product with a valid PSD kernel  $k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^\top \phi(\mathbf{x}_j)$  yields the kernelized dual:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j),$$

subject to the same constraints. The constraints involve only the labels  $y_i$  and the per-class caps, so they are unchanged.

- (g) **[Bonus] [2 points]** What happens when we set either one of  $C_{-1}$  and  $C_1$  to 0? Clearly describe the nature of the optimal solution.

**Solution:** If  $C_{-1} = 0$ : slack variables for negative examples ( $y_i = -1$ ) are unpenalized, so all violations on the negative class are free. The optimum is degenerate—negative examples impose no effective constraint, and the classifier can collapse to predicting the positive class everywhere (e.g.,  $\mathbf{w} = \mathbf{0}$ ,  $b = 1$  satisfies all positive-example constraints with  $\xi_i = 0$ , while negative-example violations are absorbed by free slack).

Symmetrically, if  $C_1 = 0$ , positive-example violations are free and the classifier can collapse to always predicting the negative class (e.g.,  $\mathbf{w} = \mathbf{0}$ ,  $b = -1$ ). In both cases the dual is also degenerate: the upper bound on the corresponding  $\alpha_i$  becomes 0, forcing those  $\alpha_i$  to be exactly 0, i.e., those points cannot be support vectors.

- (h) **[Bonus] [3 points]** How could formulation (9) be useful? For example, consider a machine learning algorithm designed to detect a disease in individuals.

**Solution:** This formulation is useful when the two classes have asymmetric costs or when the dataset is class-imbalanced. In disease detection, let  $y = +1$  mean “has disease” and  $y = -1$  mean “healthy.” A false negative (missing a sick patient) is typically far more costly than a false positive (an extra follow-up test). Choosing  $C_1 \gg C_{-1}$  penalizes margin violations on diseased patients much more heavily, encouraging the classifier to err toward catching positive cases.

Class imbalance is a related case: if positives are rare, increasing  $C_1$  relative to  $C_{-1}$  prevents the classifier from trivially predicting the majority (negative) class. The same idea underlies common “class\_weight” arguments in SVM libraries.